Continuous functions – 2

Monotonicity, oscillation and the horizontal chord

Homework

1 сентября 2026 года

Theoretical minimum. The horizontal chord theorem and the difference trick $g(x) = f(x+\alpha) - f(x)$; continuity together with injectivity implies strict monotonicity; oscillation at a point; the range of a continuous function on an interval; iterates of a continuous self-map of a segment; local extrema and monotonicity.

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1. Define the oscillation of $f\colon \mathbb{R} \to \mathbb{R}$ at x_0 as the infimum, over all neighbourhoods U of x_0, of $\sup_U f - \inf_U f$. Suppose $\operatorname{osc} f(x) = 1$ for every $x \neq 0$. Find all possible values of $\operatorname{osc} f(0)$.

2. Does there exist a continuous $f\colon \mathbb{R} \to \mathbb{R}$ such that $\lim_{x\to x_0} \frac{|f(x)-f(x_0)|}{|x-x_0|} = +\infty$ at every point x_0?

3. (IMC 2006) Let $f\colon \mathbb{R} \to \mathbb{R}$. Prove or disprove:

   a) if f is continuous and $\operatorname{range}(f) = \mathbb{R}$, then f is monotone;

   b) if f is monotone and $\operatorname{range}(f) = \mathbb{R}$, then f is continuous;

   c) if f is monotone and continuous, then $\operatorname{range}(f) = \mathbb{R}$.

4. (IMC 1996) Prove that for a continuous $f\colon [0,1] \to [0,1]$ the sequence of iterates $x_{n+1} = f(x_n)$ converges if and only if $x_{n+1} - x_n \to 0$.

5. (IMC 2023) Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ with a continuous second derivative for which $f(7x+1) = 49f(x)$ for all $x \in \mathbb{R}$.

6. (IMC 1997) Say that a continuous $f\colon [0,1] \to \mathbb{R}$ *crosses the axis* at x if $f(x) = 0$ while every neighbourhood of x contains y, z with $f(y) < 0 < f(z)$.

a) Give an example of a continuous function crossing the axis infinitely often.

b) Can a continuous function cross the axis uncountably often?

7. (RMO 2002) Let $f\colon \mathbb{R} \to \mathbb{R}$ have a limit at every point and no local extremum. Show that a) f is continuous; b) f is strictly monotone.

8. (RMO 2005) Let $f\colon [0,1) \to (0,1)$ be continuous and surjective.

   a) Prove that for every $a \in (0,1)$ the restriction $f|_{(a,1)}\colon (a,1) \to (0,1)$ is surjective.

   b) Give an example of such a function.

9. (RMO 2006) Let $f\colon [0,\infty) \to \mathbb{R}$ be such that for every $x > 0$ the sequence $\{f(nx)\}_{n\geqslant 0}$ is increasing.

   a) If f is continuous on $[0,1]$, must f be increasing?

   b) The same question if f is continuous on $[0,\infty)$.

10. (RMO 2007) Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and let $a < b$ be two points of the image of f. Show that there is a closed interval I with $f(I) = [a,b]$.